

6.1 Populations grow and decline

- 1 What is population momentum?
- 2 Fill in the blanks with the correct word(s):
 - a The number of live births per 1000 people in a year is called the _____ rate.
 - b When the death rate is higher than the birth rate, a population _____.
 - c Net migration is the balance between _____ and _____ numbers for a particular country.
- 3 Decide if the following statements are true or false:
 - a The Demographic Transition Model (DTM) has four stages.
 - b High death rates are common in countries in Stage 1 of the DTM.
 - c A youthful population is associated with high fertility rates.
- 4
 - a Describe the challenges that a country in Stage 2 of the DTM may face.
 - b What different challenges could a country in Stage 4 of the DTM face?
- 5 What is the main difference between a pro-natalist and an anti-natalist policy?
- 6 What is one strength and one weakness of the DTM?

Development and population are closely linked. As a country's level of development changes, it affects birth rates, life expectancy and access to resources. See if you can answer the questions below from [Topic 8, 'The challenge of development'](#).

- a** What is the difference between Gross Domestic Product (GDP) and Gross National Income (GNI)?
- b** Why are birth rates and death rates not reliable indicators of development when used on their own?
- c** How can international aid help reduce the development gap?

When answering questions that ask you to 'calculate', ensure you show all your working clearly and include appropriate units in your final answer (if appropriate).

- 7** Using Table 6.1, calculate the natural increase for that country. You may give your answer as a fraction or a percentage. [2]

Crude birth rate	16
Crude death rate	10

Table 6.1: Measures of fertility and mortality for a country, per thousand per year

- 8** Table 6.2 shows world population growth over time.

Year	World population (billions)
1800	1
1927	2

Year	World population (billions)
1960	3
1974	4
1987	5
1999	6
2011	7
2022	8

Table 6.2: World population growth over time

- a** Calculate how long it took to reach 2 billion people from the year 1800. [2]
- b** Calculate how long it took to reach 8 billion people from the year 2011. [2]

2 A country's birth rate was 32 births per 1000 people in 2000. By 2020, the birth rate had decreased to 24 births per 1000 people.

Calculate the percentage decrease in the birth rate over this period. Show your working. [2]

REFLECTION

Reflect on how you approach questions with the command word 'calculate'. Think about whether you double-check your calculations to avoid simple mathematical errors. How do you remember to include units?

UNDERSTAND THESE TERMS

- Population momentum
- Crude birth rate (CBR)
- Fertility rate
- Crude death rate (CDR)
- Natural increase
- Net migration
- Pro-natalist policy
- Anti-natalist policy